Laboratory Record

of _____

Roll No.: _____
Experiment No.: _____
Sheet No.: _____
Date.: _____

Aim

Build a weather forecast dashboard using JavaScript fetch API to consume a public weather API.

Description

A weather dashboard that fetches current conditions using the fetch API. It uses Leaflet.js for an interactive map, Open-Meteo for weather data and geocoding, and Nominatim to identify clicked map locations.

Algorithm

1. Get User Input: Capture the city name entered by the user in the dashboard interface.
2. Fetch Coordinates (Geocoding): Use the JavaScript fetch API to send the city name to a public API and retrieve its latitude and longitude.
3. Fetch Weather Data: Use the fetch API to pass those coordinates to a public weather API and retrieve the JSON weather forecast data.
4. Parse the Response: Extract the necessary weather details (such as temperature, wind speed, and humidity) from the returned JSON object.
5. Update Dashboard: Inject the extracted data into the HTML DOM to display the weather forecast to the user.

Program Code

Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Weather Dashboard</title>
    <link rel="stylesheet" href="https://unpkg.com/leaflet@1.9.4/dist/leaflet.css">
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <div class="container">
        <h1>Weather Dashboard</h1>
```

```
    <div id="map"></div>
    <div class="search-box">
      <input id="cityInput" placeholder="Enter city name">
      <button id="searchBtn">Search</button>
    </div>
    <p id="message"></p>
    <div id="weather" class="hidden">
      <h2 id="location"></h2>
      <div class="temp"><span id="temperature"></span>°C</div>
      <p id="description"></p>
      <p>Humidity: <span id="humidity"></span>% | Wind: <span id="wind"></span> km/h</p>
    </div>
  </div>
  <script src="https://unpkg.com/leaflet@1.9.4/dist/leaflet.js"></script>
  <script src="script.js"></script>
</body>
</html>
```

Style.css

```
body { font-family: sans-serif; background: #e0f7fa; padding: 20px; margin: 0; }
.container { max-width: 600px; margin: auto; background: white; padding: 20px; border-radius: 8px; text-align: center; }
#map { height: 300px; margin-bottom: 20px; border-radius: 8px; }
.search-box { display: flex; gap: 10px; margin-bottom: 10px; }
input { flex: 1; padding: 10px; border: 1px solid #ccc; border-radius: 4px; font-size: 16px; }
button { padding: 10px 20px; background: #007bff; color: white; border: none; border-radius: 4px; cursor: pointer; }
.hidden { display: none; }
.temp { font-size: 40px; font-weight: bold; color: #007bff; margin: 10px 0; }
```

Script.js

```
const $ = id => document.getElementById(id);

const map = L.map("map").setView([20, 0], 2);

L.tileLayer("https://{s}.tile.openstreetmap.org/{z}/{x}/{y}.png").addTo(map);

let marker;

const setWeather = async (lat, lng, name) => {
  try {
    const res = await fetch(`https://api.open-meteo.com/v1/forecast?latitude=${lat}&longitude=${lng}&current=temperature_2m,relative_humidity_2m,wind_speed_10m,weather_code`);
    const { current: c } = await res.json();
    $("location").textContent = name;
    $("temperature").textContent = Math.round(c.temperature_2m);
    $("humidity").textContent = c.relative_humidity_2m;
    $("wind").textContent = c.wind_speed_10m;
    const wCodes = {0:"Clear", 1:"Mainly clear", 2:"Partly cloudy", 3:"Overcast", 45:"Fog", 51:"Drizzle", 61:"Rain", 71:"Snow", 95:"Thunderstorm"};
    $("description").textContent = wCodes[c.weather_code] || "Unknown";
    $("weather").classList.remove("hidden");
    $("message").textContent = "";
    if (marker) map.removeLayer(marker);
    marker = L.marker([lat, lng]).addTo(map).bindPopup(`<b>${name}</b>`).openPopup();
  } catch {
    $("message").textContent = "Error fetching weather.";
  }
};

const searchCity = async () => {
  const city = $("cityInput").value.trim();
  if (!city) return $("message").textContent = "Enter a city.";
  $("message").textContent = "Loading...";
```

```
  $("weather").classList.add("hidden");
  try {
    const res = await fetch(`https://geocoding-api.open-meteo.com/v1/search?name=${city}&count=1&format=json`);
    const { results } = await res.json();
    if (!results) throw new Error();
    const { latitude: lat, longitude: lng, name, country } = results[0];
    map.setView([lat, lng], 10);
    setWeather(lat, lng, `${name}, ${country}`);
  } catch {
    $("message").textContent = "City not found.";
  }
};
$("searchBtn").addEventListener("click", searchCity);
$("cityInput").addEventListener("keyup", e => {
  if (e.key === "Enter") {
    searchCity();
  }
});
map.on("click", async e => {
  const { lat, lng } = e.latlng;
  $("message").textContent = "Finding location...";
  $("weather").classList.add("hidden");
  try {
    const res = await fetch(`https://nominatim.openstreetmap.org/reverse?lat=${lat}&lon=${lng}&format=json`);
    const { address: addr } = await res.json();
    const name = addr?.city || addr?.town || addr?.village || addr?.country || "Selected Location";
    setWeather(lat, lng, name);
```

```
  } catch {
      $("message").textContent = "Location not found.";
  }
});
```

